data sets with hundreds and thousands of items, whether text or images. To look at some metrics, the cost of training GPT-3 is over 4.6 million dollars using a Tesla V100 cloud instance with training times of up to nine days. Similarly, Stable Diffusion was trained using 256 Nvidia A100 GPUs on Amazon Web Services for a total of 150,000 GPU hours, at a cost of US$ 600,000. These costs are prohibitive for individuals and even small businesses. Putting these trained models to use is more inexpensive by orders of magnitude.

Fortunately, it is possible to share the outcome of this training in the form of 'weights', which can be loaded by anyone who wants to use the model. This is the approach taken by GPT-J 6B and Stable Diffusion, who had weights available for download from day one. This allows individuals that have access to sufficiently powerful hardware to load up the model and put it to use.

## Highly efficient architecture with lower hardware requirements for deployment

Even after obtaining a set of weights to pretrain your model, many of them require expensive GPUs like the Nvidia A100. The primary resource essential here is large reserves of VRAM, in the range of 10s of gigabytes. The VRAM is required to keep model weights in memory for reasonably fast execution. A requirement like this automatically filters out individuals with consumer-grade, yet fairly powerful GPUs.

For instance, running the full-fat version of GPT-J-6B requires upwards of 20-25GB of VRAM. Additionally, in many practical use cases, you would want to "fine-tune" the model, which involves making the model more specialised to a given use case. This is done by retraining the already trained model over a specific, curated data set. This is yet another process that requires powerful GPUs.

Stable Diffusion gets past this by utilising a novel architecture known as 'latent diffusion', which maps the image generation process to a smaller logical space. This allows it to run with reasonable performance on GPUs with even 8GB of VRAM, a figure easily attainable in consumer GPUs. This meant that, on release, anyone with a laptop with reasonable specifications could run the model entirely locally, resulting in no hard limits imposed on the model in terms of output filtering or the rate at which one could iterate.

You don't need to be in the space of AI/ML to be excited by these advances. Generative AI is penetrating our lives and making us more efficient. As a developer and open source maintainer myself, I find myself using generative AI in my day-to-day tasks.

I have a significant backlog of articles that I've always wanted to read and gain insights from. I now find myself using ChatGPT to summarise and ask introspective questions about those articles. In the same vein, I also use it to refine and refactor code documentation, READMEs, and other artefacts in a pinch. I've also extensively used GitHub Copilot for writing boilerplate code and exploring unfamiliar codebases.

From a more creative perspective, I've begun to extensively use Stable Diffusion to provide abstract imagery for my presentations and to edit images to make minor modifications. As an experiment, I've been using a fine-tuned GPT-3 model and Stable Diffusion to aid my poetry writing.

In my use of generative AI, I've found that it's currently far away from entirely replacing any sort of human-generated work. However, it is an invaluable asset in anyone's toolbox that enables us to do more with less. It enables non-native speakers to write effectively, technologists to rapidly consume and summarise knowledge, and brings iterative artistry to creatives and even creatively challenged individuals. In a way, it's open sourcing creativity!

It's important to remember that just like other disruptive technologies, generative AI has its downsides. One of the most contentious issues is around the legality and ethical morality of using public content as training data. It's been argued that AI shouldn't be allowed to create derived works from the content, especially since current-generation models are incapable of appropriate attribution. From a FOSS standpoint, there has been a heavy backlash around GitHub's use of code in open repositories to train Copilot, a paid product.

Additionally, as we employ generative AI, we must be cognizant that the models only possess an abstract understanding inferred from the training data. This means that we can't know for sure if they are giving us the right answers. This becomes incredibly obvious when you try pushing text-based models to generate content around items that aren't present in their training set. Most models confidently generate sentences that seem correct from a high level but are entirely inaccurate. This has led to the ban of AI-generated responses from popular Q&A sites like Stack Overflow.

In conclusion, it's an exciting time for this field! And I for one cannot wait to see where we go in the future.

*P.S.: Staying true to the spirit of 'practice what you preach', this article has been co-edited by ChatGPT!* END

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The author is a senior developer specialising in emerging technologies. He started his journey with open source at the age of 12 and has never looked back since!